

## Solutions of Tutorial #5

### Topics covered:

- The product and the box topologies
- The metric topology

### Problem 1

Show that  $(X_1 \times \cdots \times X_{n-1}) \times X_n$  is homeomorphic to  $X_1 \times \cdots \times X_n$ .

**Solution:** First we note that since we are dealing with finite products, the box and product topologies are the same; we shall find it most convenient to use the box topology definition. Also, as there are no intervals involved here, we use the traditional tuple notation using parentheses. So define  $f : X_1 \times \cdots \times X_n \rightarrow (X_1 \times \cdots \times X_{n-1}) \times X_n$  by

$$f(x_1, \dots, x_n) = ((x_1, \dots, x_{n-1}), x_n).$$

It is obvious that this is a bijection, and it is trivial to prove. Also obvious and trivial to prove based on the definition of  $f$  is that  $f(A_1 \times \cdots \times A_n) = (A_1 \times \cdots \times A_{n-1}) \times A_n$  when each  $A_k \subset X_k$ .

First, we show that  $f$  is continuous by showing that the inverse image of every basis element in  $(X_1 \times \cdots \times X_{n-1}) \times X_n$  is open in  $X_1 \times \cdots \times X_n$ . So consider any basis element  $C$  of  $(X_1 \times \cdots \times X_{n-1}) \times X_n$  and let  $U = f^{-1}(C)$  so that of course  $f(U) = C$  and  $U \subset X_1 \times \cdots \times X_n$ . We then have that  $C = V' \times V_n$  where  $V'$  is open in  $X_1 \times \cdots \times X_{n-1}$  and  $V_n$  is open in  $X_n$  by the definition of the box/product topology. Now consider any  $x \in U$  so that  $x = (x_1, \dots, x_n)$  and we have that  $f(x) = ((x_1, \dots, x_{n-1}), x_n) \in f(U) = C$ . Hence  $x' = (x_1, \dots, x_{n-1}) \in V'$  and  $x_n \in V_n$ . Since  $V'$  is open in  $X_1 \times \cdots \times X_{n-1}$  there is a basis element  $C'$  containing  $x'$  that is a subset of  $V'$ . By the definition of the box topology, we then have that  $C' = V_1 \times \cdots \times V_{n-1}$  where each  $V_k$  is open in  $X_k$ .

We then have that  $B = V_1 \times \cdots \times V_n$  is a basis element of  $X_1 \times \cdots \times X_n$  and also clearly  $B$  contains  $x$  since  $(x_1, \dots, x_{n-1}) = x' \in C' = V_1 \times \cdots \times V_{n-1}$  and  $x_n \in V_n$ . Now suppose that  $y = (y_1, \dots, y_n) \in B$  so that each  $y_k \in V_k$ . Then we have that  $y' = (y_1, \dots, y_{n-1}) \in C'$  so that also  $y' \in V'$  since  $C' \subset V'$ . Since also of course  $y_n \in V_n$ , we have that  $(y', y_n) \in V' \times V_n = C$ . Also clearly  $f(y) = (y', y_n) \in C = f(U)$  so that  $y \in U$ . Since  $y$  was arbitrary this shows that  $B \subset U$ , which suffices to show that  $U$  is open since  $x$  was arbitrary. This completes the proof that  $f$  is continuous. We based on the definition of  $f$  is that  $f(A_1 \times \cdots \times A_n)$  First we show that  $f$  is continuous by showing that  $(X_1 \times \cdots \times X_{n-1}) \times X_n$  is open in  $X_1 \times \cdots \times X_n$ .

and let  $U = f^{-1}(C)$  so that of course have that  $C = V' \times V_n$  where  $V'$  is open in  $X_1 \times \cdots$  of the box / product topology. Now consider any  $x \in f(U) = ((x_1, \dots, x_{n-1}), x_n) \in f(U) = C$ . Hence  $x'$  is open in  $X_1 \times \cdots \times X_{n-1}$  there is a basis element definition of the box topology, we then have that  $C'$  We then have that  $B = V_1 \times \cdots \times V_n$  is a basis contains  $x$  since  $(x_1, \dots, x_{n-1}) = x' \in C' = V_1 \times \cdots \times V_{n-1}$  so that each  $y_k \in V_k$ . Then we  $y' \in V'$  since  $C' \subset V'$ . Since also of course  $y_n \in V$  clearly  $f(y) = (y', y_n) \in C = f(U)$  so that  $y \in U$ . which suffices to show that  $U$  is open since  $x$  was continuous.

Next we show that  $f^{-1}$  is continuous, which is a little simpler. Let  $B$  be any basis element of  $X_1 \times \cdots \times X_n$  so that  $B = U_1 \times \cdots \times U_n$  where each  $U_k$  is open in  $X_k$  by the definition of the box topology. Then we have that  $f(B) = (U_1 \times \cdots \times U_{n-1}) \times U_n$ . By the definition of the box topology, we then have that  $U' = U_1 \times \cdots \times U_{n-1}$  is a basis element of  $X_1 \times \cdots \times X_{n-1}$  and is therefore open. Since  $U_n$  is also open, we have that  $f(B) = U' \times U_n$  is a basis element of  $(X_1 \times \cdots \times X_{n-1}) \times X_n$  by the definition of the box/product topology, and is therefore open. Since  $f(B) = (f^{-1})^{-1}(B)$  is the inverse image of  $B$  under  $f^{-1}$ , this shows that  $f^{-1}$  is also continuous. We have shown that both  $f$  and  $f^{-1}$  are continuous, which proves that  $f$  is a homeomorphism by definition. ■

## Problem 2

In the lectures we proved that given  $f : A \rightarrow \prod_{\alpha \in J} X_\alpha$  by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J},$$

where  $f_\alpha : A \rightarrow X_\alpha$  for each  $\alpha$  and  $\prod X_\alpha$  is endowed with the product topology, then the function  $f$  is continuous if and only if each function  $f_\alpha$  is continuous. One of the two implications in this equivalence holds for the box topology. Which one? Prove it.

**Solution:** Also in the lectures we provided an example of a function  $f$  that is not continuous in the box topology even though all of its component functions  $f_\alpha$  are continuous. Hence the only implication that can be generally true in the box topology is that  $f$  being continuous implies that each  $f_\alpha$  is continuous. A proof of this is straightforward.

Suppose that  $f : A \rightarrow \prod_{\alpha \in J} X_\alpha$  be given by

$$f(a) = (f_\alpha(a))_{\alpha \in J},$$

where  $f_\alpha : A \rightarrow X_\alpha$  for each  $\alpha \in J$ . Here  $\prod X_\alpha$  has the box topology. Suppose that  $f$  is continuous and consider any  $\beta \in J$ . We show that  $f_\beta$  is continuous, which of course shows the desired result. So let  $V$  be any open set of  $X_\beta$  and define

$$B_\alpha = \begin{cases} V & \alpha = \beta \\ X_\alpha & \alpha \neq \beta \end{cases}$$

Then, since each  $B_\alpha$  is clearly open in  $X_\alpha$ , we have that  $B = \prod B_\alpha$  is a basis element of the box topology by definition and is therefore open. Hence  $U = f^{-1}(B)$  is open in  $A$  since  $f$  is continuous.

We claim that  $U = f_\beta^{-1}(V)$ , which shows that  $f_\beta$  is continuous since  $U$  is open in  $A$  and  $V$  was an arbitrary open set of  $X_\beta$ . We prove it by double inclusion:

( $\subset$ ) If  $x \in U = f^{-1}(B)$  then of course  $f(x) \in B$  so that each  $f_\alpha(x) \in B_\alpha$  since  $f(x) = (f_\alpha(x))_{\alpha \in J}$  and  $B = \prod B_\alpha$ . In particular  $f_\beta(x) \in B_\beta = V$  so that  $x \in f_\beta^{-1}(V)$ . Hence  $U \subset f_\beta^{-1}(V)$  since  $x$  was arbitrary.

( $\supset$ ) If  $x \in f_\beta^{-1}(V)$  then  $f_\beta(x) \in V = B_\beta$ . Since of course every other  $f_\alpha(x) \in X_\alpha = B_\alpha$  we have that  $f(x) \in \prod B_\alpha = B$ . Hence  $x \in f^{-1}(B) = U$  so that  $f_\beta^{-1}(V) \subset U$  since  $x$  was arbitrary. ■

### Problem 3

Let  $\mathbb{R}^\infty$  be the subset of  $\mathbb{R}^\omega$  consisting of all sequences that are “eventually zero”, that is, all sequences  $(x_1, x_2, \dots)$  such that  $x_i \neq 0$  for only finitely many values of  $i$ . What is the closure of  $\mathbb{R}^\infty$  in  $\mathbb{R}^\omega$  in the box and product topologies? Justify your answer.

**Solution:** First, we claim that  $\mathbb{R}^\infty$  is dense in  $\mathbb{R}^\omega$  in the product topology in the sense that its closure is all of  $\mathbb{R}^\omega$ :

We show that any point of  $\mathbb{R}^\omega$  is in  $\overline{\mathbb{R}^\infty}$ . So consider any point  $x = (x_1, x_2, \dots) \in \mathbb{R}^\omega$  and any neighborhood  $U$  of  $x$ . Then there is a basis element  $B = \prod U_n$  containing  $x$  where  $B \subset U$ . By the definition of the product topology each  $U_n$  is open and  $U_n = \mathbb{R}$  for all but finitely many values of  $n$ . So let  $I$  be a finite subset of  $\mathbb{Z}_+$  such that  $U_n = \mathbb{R}$  for all  $n \notin I$  and  $U_n$  is merely just open for  $n \in I$ . Consider now the sequence  $y = (y_1, y_2, \dots)$  defined by

$$y_n = \begin{cases} x_n & n \in I \\ 0 & n \notin I \end{cases}$$

for  $n \in \mathbb{Z}_+$ . Since  $I$  is finite clearly  $y \in \mathbb{R}^\infty$ . Also  $y_n = x_n \in U_n$  when  $n \in I$  since  $B = \prod U_n$  contains  $x$ . We also have  $y_n = 0 \in \mathbb{R} = U_n$  when  $n \notin I$  so that either way  $y_n \in U_n$  and hence  $y \in \prod U_n = B$ . Thus also  $y \in U$  since  $B \subset U$ . Since  $U$  was an arbitrary neighborhood and  $U$  intersects  $\mathbb{R}^\infty$  (with  $y$  being a point in the intersection), this shows that  $x \in \overline{\mathbb{R}^\infty}$  by Theorem 17.5. This of course shows the desired result since  $x$  was any element of  $\mathbb{R}^\omega$ .

For the box topology, we claim that  $\mathbb{R}^\infty$  is already closed:

We show this by proving that any point not in  $\mathbb{R}^\infty$  is not a limit point of  $\mathbb{R}^\infty$  so that  $\mathbb{R}^\infty$  must already contain all its limit points. So consider any  $x = (x_1, x_2, \dots) \notin \mathbb{R}^\infty$  so that  $x_n \neq 0$  for infinitely many values of  $n$ . Now define the sets

$$U_n = \begin{cases} (-1, 1) & x_n = 0 \\ (x_n/2, 2x_n) & x_n > 0 \\ (2x_n, x_n/2) & x_n < 0 \end{cases}$$

for  $n \in \mathbb{Z}_+$ . Clearly each  $U_n$  is a basis element of  $\mathbb{R}$  and is therefore open. Also clearly each  $x_n \in U_n$ . It therefore follows that  $B = \prod U_n$  is a basis element of  $\mathbb{R}^\omega$  and is therefore open,

and that  $x \in B$ . Hence  $B$  is a neighborhood of  $x$ . Then, for any  $y = (y_1, y_2, \dots) \in B$  we have that each  $y_n \in U_n$ . For infinitely many  $n \in \mathbb{Z}_+$  we then have that  $x_n \neq 0$  and hence  $x_n > 0$  or  $x_n < 0$ . In the former case  $y_n \in U_n = (x_n/2, 2x_n)$  so that  $0 < x_n/2 < y_n$ . In the latter case  $y_n \in U_n = (2x_n, x_n/2)$  so that  $y_n < x_n/2 < 0$ . Hence either way  $y_n \neq 0$  so that  $y \notin \mathbb{R}^\infty$  since this is true for infinitely many  $n$ . Since  $y \in B$  was arbitrary, this shows that  $B$  cannot not intersect  $\mathbb{R}^\infty$ . Therefore  $x$  is not a limit point of  $\mathbb{R}^\infty$  since  $B$  is a neighborhood of  $x$ . ■

#### Problem 4

Let  $X$  be metric space with metric  $d$ .

- (a) Show that  $d : X \times X \rightarrow \mathbb{R}$  is continuous when  $\mathbb{R}$  is endowed with the standard topology.
- (b) Let  $X'$  denote a topological space having the same underlying set as  $X$  but with a topology  $\mathcal{T}'$  which is not necessarily the metric topology induced by  $d$ .

Show that if  $d : X' \times X' \rightarrow \mathbb{R}$  is continuous, then the topology of  $\mathcal{T}'$  is finer than the metric topology of  $X$ . One can summarize the result of this exercise as follows: If  $X$  has a metric  $d$ , then the topology induced by  $d$  is the coarsest topology relative to which the function  $d$  is continuous.

**Solution:** (a) We use Theorem 18.1 part (4) to show that  $d$  is continuous. So consider any  $(x_1, x_2) \in X \times X$  and any neighborhood  $V$  of  $z = d(x_1, x_2)$ , noting that  $V \subset \mathbb{R}$  since  $\mathbb{R}$  is the range of  $d$ . Since  $V$  is open in  $\mathbb{R}$ , there is a basis element  $B = (a, b)$  containing  $z$  where  $B \subset V$ . Hence  $a < z < b$ . Now let  $\epsilon = \min\{(z - a)/2, (b - z)/2\}$ , noting that  $\epsilon > 0$  since  $z > a$  and  $b > z$ . Next define  $U_1 = B_d(x_1, \epsilon)$  and  $U_2 = B_d(x_2, \epsilon)$  so that they are both basis elements and therefore open sets of the metric space  $X$ . It then follows that  $U = U_1 \times U_2$  is a basis element and therefore an open set of the product space  $X \times X$ . Clearly we have that  $x_1 \in B_d(x_1, \epsilon) = U_1$  and  $x_2 \in B_d(x_2, \epsilon) = U_2$  so that  $U$  contains  $(x_1, x_2)$  and so is a neighborhood of  $(x_1, x_2)$ .

We claim that  $d(U) \subset B$ . To see this, consider any  $w \in d(U)$  so that there is a  $y_1 \times y_2 \in U = U_1 \times U_2$  such that  $w = d(y_1, y_2)$ . Therefore  $y_1 \in U_1 = B_d(x_1, \epsilon)$  so that  $d(y_1, x_1) < \epsilon$ , and similarly  $d(y_2, x_2) < \epsilon$  since  $y_2 \in U_2 = B_d(x_2, \epsilon)$ . Then, since  $d$  is a metric, we have

$$\begin{aligned}
 z = d(x_1, x_2) &\leq d(x_1, y_1) + d(y_1, x_2) \\
 &\leq d(x_1, y_1) + d(y_1, y_2) + d(y_2, x_2) \\
 &= d(y_1, x_1) + d(y_1, y_2) + d(y_2, x_2) \\
 &< \epsilon + w + \epsilon \\
 z &< w + 2\epsilon \leq w + 2(z - a)/2 = w + z - a \\
 a &< w
 \end{aligned}$$

Similarly, we have

$$\begin{aligned}
w = d(y_1, y_2) &\leq d(y_1, x_1) + d(x_1, y_2) \\
&\leq d(y_1, x_1) + d(x_1, x_2) + d(x_2, y_2) \\
&= d(y_1, x_1) + d(x_1, x_2) + d(y_2, x_2) \\
&< \epsilon + z + \epsilon \\
w &< z + 2\epsilon \leq z + 2(b - z)/2 = z + b - z \\
w &< b
\end{aligned}$$

We therefore have that  $a < w < b$  so that  $w \in (a, b) = B$ . This of course shows that  $d(U) \subset B$  since  $w$  was arbitrary. Moreover, we have that  $B \subset V$  so that clearly  $d(U) \subset V$ , which completes the proof of Theorem 18.1 part (4) so that  $d$  is continuous.

(b) Let  $U$  be any open set of  $X$  and consider any  $x \in U$ . Then clearly there is a basis element  $B_d(y, \epsilon)$ , for some  $\epsilon > 0$  and  $y \in U$ , of the metric topology  $X$  that contains  $x$  and where  $B_d(y, \epsilon) \subset U$ . Now, since  $d$  is continuous with respect to  $X' \times X'$ , it follows from Problem 8 in Tutorial 4 that the function  $d_y(z) = d(y, z)$  is a continuous function from  $X'$  to  $\mathbb{R}$ . Since clearly the interval  $(-\infty, \epsilon)$  is open in  $\mathbb{R}$ , it then follows that the set

$$d_y^{-1}((-\infty, \epsilon)) = \{z \in X' \mid d_y(z) < \epsilon\} = \{z \in X \mid d(y, z) < \epsilon\} = B_d(y, \epsilon)$$

is also open in  $X'$ . Thus  $B_d(y, \epsilon)$  is an open set in  $X'$  containing  $x$  such that  $B_d(y, \epsilon) \subset U$ . This shows that  $U$  is also open in  $X'$  since the point  $x \in U$  was arbitrary. This suffices to show the desired result. ■

## Problem 5

Given the points  $\mathbf{x} = (x_\alpha)_{\alpha \in \mathbb{N}}$  and  $\mathbf{y} = (y_\alpha)_{\alpha \in \mathbb{N}}$  of  $\mathbb{R}^\omega$ , let us define a metric  $\bar{\rho}$  on  $\mathbb{R}^\omega$  by the equation

$$\bar{\rho}(\mathbf{x}, \mathbf{y}) = \sup \{ \bar{d}(x_\alpha, y_\alpha) \mid \alpha \in \mathbb{N} \},$$

where  $\bar{d}$  is the standard bounded metric on  $\mathbb{R}$ . It is easy to check that  $\bar{\rho}$  is indeed a metric; it is called the **uniform metric** on  $\mathbb{R}^\omega$ , and the topology it induces is called the **uniform topology**. In Theorem 20.4 in the textbook it is proved that the uniform topology on  $\mathbb{R}^\omega$  is finer than the product topology and coarser than the box topology; these three topologies are all different.

Let now  $\mathbb{R}^\infty$  be the subset of  $\mathbb{R}^\omega$  consisting of all sequences that are eventually zero (see Problem 3). What is the closure of  $\mathbb{R}^\infty$  in  $\mathbb{R}^\omega$  in the uniform topology? Justify your answer.

**Solution:** Let  $\mathbb{R}^0$  denote the subset of  $\mathbb{R}^\omega$  consisting of all sequences that converge to zero. We then claim that  $\overline{\mathbb{R}^\infty} = \mathbb{R}^0$ , i.e. the closure of  $\mathbb{R}^\infty$  is  $\mathbb{R}^0$ .

( $\subset$ ) We show this by contrapositive. So suppose that  $\mathbf{x} \notin \mathbb{R}^0$  so that the sequence  $\mathbf{x}$  does not converge to zero. Then there is a neighborhood  $V$  of 0 in the standard topology on  $\mathbb{R}$  such that, for any  $N \in \mathbb{Z}_+$ , there is an  $n \geq N$  where  $x_n \notin V$ . It also follows from Lemma

20.4.1 that there is an  $\epsilon > 0$  such that  $B_d(0, \epsilon) \subset V$ . So let  $\delta = \min\{\epsilon, 1\}$  and consider the set  $B_{\bar{\rho}}(\mathbf{x}, \delta)$ , which is clearly a neighborhood of  $\mathbf{x}$  in the uniform topology. Also suppose that  $\mathbf{y}$  is any element of  $\mathbb{R}^\infty$  so that  $\mathbf{y}$  is eventually zero. Then there must be an  $N$  where  $y_n = 0$  for all  $n \geq N$ . From before we have that there is a specific  $n \geq N$  where  $x_n \notin V$  so that also  $x_n \notin B_d(0, \epsilon)$ , and thus

$$d(y_n, x_n) = d(0, x_n) = d(x_n, 0) \geq \epsilon \geq \delta.$$

Then we have that both  $d(y_n, x_n) \geq \delta$  and  $1 \geq \delta$  so that

$$\bar{d}(y_n, x_n) = \min\{d(y_n, x_n), 1\} \geq \delta.$$

From this it clearly follows that

$$\bar{\rho}(\mathbf{y}, \mathbf{x}) = \sup\{\bar{d}(y_n, x_n) \mid n \in \mathbb{Z}_+\} \geq \delta$$

so that  $\mathbf{y} \notin B_{\bar{\rho}}(\mathbf{x}, \delta)$ . Since  $\mathbf{y}$  was arbitrary, this shows that  $B_{\bar{\rho}}(\mathbf{x}, \delta)$  does not intersect  $\mathbb{R}^\infty$ . This in turn shows that  $\mathbf{x}$  is not a limit point of  $\mathbb{R}^\infty$ . Now, clearly also  $\mathbf{x}$  cannot be eventually zero since then it would converge to zero, hence  $\mathbf{x} \notin \mathbb{R}^\infty$  either. Therefore  $\mathbf{x}$  cannot be in the closure of  $\mathbb{R}^\infty$ . Hence by the contrapositive we have that  $\overline{\mathbb{R}^\infty} \subset \mathbb{R}^0$ .

( $\supset$ ) Now consider any  $\mathbf{x} \in \mathbb{R}^0$  so that the sequence  $\mathbf{x}$  converges to zero. Consider any neighborhood  $U$  of  $\mathbf{x}$  in the uniform topology so that, by Lemma 20.4.1, there is an  $\epsilon > 0$  such that  $B_{\bar{\rho}}(\mathbf{x}, \epsilon) \subset U$ . Now, since  $B_d(0, \epsilon/2)$  is a neighborhood of 0, it follows that there is an  $N \in \mathbb{Z}_+$  where  $x_n \in B_d(0, \epsilon/2)$  for all  $n \geq N$  since  $\mathbf{x}$  converges to 0. Now define the sequence  $\mathbf{y}$  where

$$y_n = \begin{cases} x_n & n < N \\ 0 & n \geq N \end{cases}$$

for  $n \in \mathbb{Z}_+$ . Clearly  $\mathbf{y}$  is eventually zero so that  $\mathbf{y} \in \mathbb{R}^\infty$ . We also claim that  $\mathbf{y} \in U$ . To see this consider any  $n \in \mathbb{Z}_+$ . If  $n < N$  then clearly  $y_n = x_n$  so that

$$\bar{d}(y_n, x_n) \leq d(y_n, x_n) = d(x_n, x_n) = 0 < \epsilon/2.$$

If  $n \geq N$  then  $y_n = 0$  and we have from before that  $x_n \in B_d(0, \epsilon/2)$  and hence

$$\bar{d}(y_n, x_n) \leq d(y_n, x_n) = d(0, x_n) = d(x_n, 0) < \epsilon/2.$$

Hence it follows that

$$\bar{\rho}(\mathbf{y}, \mathbf{x}) = \sup\{\bar{d}(y_n, x_n) \mid n \in \mathbb{Z}_+\} \leq \epsilon/2 < \epsilon$$

so that  $\mathbf{y} \in B_{\bar{\rho}}(\mathbf{x}, \epsilon) \subset U$ . This shows that  $\mathbf{y} \in \mathbb{R}^\infty \cap U$ . If  $\mathbf{y} = \mathbf{x}$  then of course  $\mathbf{x} = \mathbf{y} \in \mathbb{R}^\infty$  itself. If  $\mathbf{y} \neq \mathbf{x}$  then we have shown that  $\mathbf{x}$  is a limit point of  $\mathbb{R}^\infty$  since  $U$  was an arbitrary neighborhood. Thus either way  $\mathbf{y} \in \overline{\mathbb{R}^\infty}$  so that  $\overline{\mathbb{R}^\infty} \supset \mathbb{R}^0$  since  $\mathbf{x}$  was arbitrary.

Lastly, we note that  $\mathbb{R}^\infty$  is a proper subset of its closure  $\overline{\mathbb{R}^\infty} = \mathbb{R}^0$  since, for example, the sequence  $\mathbf{x}$  defined by  $x_n = 1/n$  for all  $n \in \mathbb{Z}_+$  clearly converges to zero so is in  $\mathbb{R}^0$  but is not eventually zero so is not in  $\mathbb{R}^\infty$ . ■

### Problem 6

Let  $A \subset X$ . If  $d$  is a metric for the topology of  $X$ , show that  $d|_{A \times A}$  is a metric for the subspace topology on  $A$ .

**Solution:** Let us denote the restricted function  $d|_{A \times A}$  by  $d'$ . Clearly  $d'$  is a metric on  $A$ , since for  $x, y \in A$  we have  $d'(x, y) = d(x, y)$  and  $d$  has all the properties of a metric. We show that the metric topology induced by  $d'$  is the same as the subspace topology using Lemma 13.3 in both directions.

First we show that  $B_{d'}(x, \epsilon) = A \cap B_d(x, \epsilon)$  for any  $x \in A$  and  $\epsilon > 0$ . For any  $y \in B_{d'}(x, \epsilon)$  we have that clearly  $y \in A$  since the metric  $d'(y, x)$  must be defined, and  $d(y, x) = d'(y, x) < \epsilon$  so that also  $y \in B_d(x, \epsilon)$ . Thus  $y \in A \cap B_d(x, \epsilon)$ , and hence  $B_{d'}(x, \epsilon) \subset A \cap B_d(x, \epsilon)$ . For the other direction suppose that  $y \in A \cap B_d(x, \epsilon)$  so that  $y \in A$  and  $y \in B_d(x, \epsilon)$ . Thus  $y$  and  $x$  are in  $A$  so that  $d'(y, x)$  is defined and  $d'(y, x) = d(y, x) < \epsilon$ , and hence  $y \in B_{d'}(x, \epsilon)$ . This shows that  $B_{d'}(x, \epsilon) \supset A \cap B_d(x, \epsilon)$ , which in turn shows that the two sets are the same.

Now, clearly each basis element of the metric topology  $B_{d'}(x, \epsilon) = A \cap B_d(x, \epsilon)$  is also a basis element of the subspace topology by Lemma 16.1. Hence for any  $x \in A$  we have that  $B_m = B_{d'}(x, \epsilon)$  is a basis element of the metric topology and  $B_s = A \cap B_d(x, \epsilon)$  is a basis element of the subspace topology. But  $B_m = B_s$  so that  $x \in B_s \subset B_m$  and  $x \in B_m \subset B_s$  so that each topology is finer than the other by Lemma 13.3. Thus they are the same topologies.

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